Asymptotic Notations-Numerical

O(Big-Oh), Ω (Big-Omega) and θ (Theta)

OBJECTIVES

After completing this section, you will be able to

- Differentiate between $O(Big - Oh), \Omega(Big - Omega)\ and\ \Theta\ (Theta)\ Notation$
- Numerical questions on $O, \Omega\ and\ \Theta$.
- Check your progress on $O, \Omega\ and\ \Theta$.

$O(1)$	$O(logn)$	$O(\sqrt{n})$	$O(n)$	$O(nlogn)$	$O(n^2)$	$O(n^3)$	…	$O(2^n)$	$O(3^n)$	$O(n^n)$

Example1:

Let $f(n) = 2n^2 + 3n + 4$ then

1) $f(n) = 2n^2 + 3n + 4 \leq 2n^2 + 3n^2 + 4n^2$

$$\leq 9n^2 \quad \rightarrow f(n) = O(n^2)$$

2) $f(n) = 2n^2 + 3n + 4 \geq 1.n^2 \quad \rightarrow f(n) = \Omega(n^2)$

3) $f(n) = 1.n^2 \leq 2n^2 + 3n^2 + 4n^2 \leq 9n^2 \rightarrow f(n) = \theta(n^2)$

$O(1)$	$O(logn)$	$O(\sqrt{n})$	$O(n)$	$O(nlogn)$	$O(n^2)$	$O(n^3)$	…	$O(2^n)$	$O(3^n)$	$O(n^n)$

Example2:

$f(n) = n^2 + n^2logn$; we can write

$$1.n^2logn \le n^2 + n^2logn \le 2.n^2logn$$

$$\to \theta(n^2logn)$$

$$\to O(n^2logn)$$

$$\to \Omega(n^2logn)$$

$O(1)$	$O(logn)$	$O(\sqrt{n})$	$O(n)$	$O(nlogn)$	$O(n^2)$	$O(n^3)$	…	$O(2^n)$	$O(3^n)$	$O(n^n)$

Example3:

$$f(n) = n! = n.(n-1).(n-2)\ldots .3.2.1$$

we can write

$$1 \times 1 \times 1 \times \cdots \times 1 \leq 1 \times 2 \times 3 \ldots . \times n \leq n \times n \times \cdots \times n$$

$$1 \leq n! \leq n^n$$

Note: Both side not getting same function, so for n! , tight bound θ not possible, so we can write $O(n^n)$ or $\Omega(1)$.

To get Theta notation, both the upper and lower bounds must have the **same order**.

$O(1)$	$O(logn)$	$O(\sqrt{n})$	$O(n)$	$O(nlogn)$	$O(n^2)$	$O(n^3)$	…	$O(2^n)$	$O(3^n)$	$O(n^n)$

Example4:

$$f(n) = \log(n!) = \log(n.(n-1).(n-2) \ldots .3.2.1)$$

we can write

$$\log(1 \times 1 \times 1 \times \cdots \times 1) \leq \log(1 \times 2 \times 3 \ldots . \times n) \leq \log(n \times n \times \cdots \times n)$$

$$1 \leq logn! \leq \log(n^n)$$

$$1 \leq logn! \leq nlogn$$

Note: **Again,** Both side not getting same function, so for $logn!$, tight bound θ not possible, so we can write $O(nlogn)$ or $\Omega(1)$.

Check your progress1:

- Example1: For the function defined by $f(n) = 2n^2 + 3n + 1$, Prove/Disprove the following:

$(i) f(n) = O(n^2)$

$(ii)\ f(n) = O(n^3)$

$(iii) n^2 = O(f(n))$

$(iv)\ f(n) \neq O(n)$

$(v)\ n^3 \neq O(f(n))$

(i) $f(n) = O(n^2)$

Definition of Big-O

We need to find positive constants c and n_0 such that

$$f(n) \le cn^2$$

for all $n \ge n_0$.

Substitute $f(n)$:

$$2n^2 + 3n + 1 \le cn^2$$

For $n \ge 1$,

$$3n \le 3n^2$$

and

$$1 \le n^2$$

Therefore,

$$2n^2 + 3n + 1 \le 2n^2 + 3n^2 + n^2$$
$$= 6n^2$$

Choose

$$c = 6, n_0 = 1$$

Hence,

$$\boxed{f(n) = O(n^2)}$$

✅ **True**

(iv) $f(n) \ne O(n)$

Suppose

$$2n^2 + 3n + 1 = O(n)$$

Then there must exist a constant c such that

$$2n^2 + 3n + 1 \le cn$$

Divide by n:

$$2n + 3 + \frac{1}{n} \le c$$

As $n \to \infty$,

$$2n + 3 + \frac{1}{n} \to \infty$$

No constant c can satisfy this.

Hence,

$$\boxed{f(n) \ne O(n)}$$

✅ **True**

(v) $n^3 \neq O\big(f(n)\big)$

We need to check whether

$$n^3 = O(2n^2 + 3n + 1)$$

Divide both sides by n^2:

$$\frac{n^3}{2n^2 + 3n + 1}$$

For large n,

$$\frac{n^3}{2n^2 + 3n + 1} \approx \frac{n}{2}$$

As $n \to \infty$,

$$\frac{n}{2} \to \infty$$

Therefore, no constant c exists such that

$$n^3 \leq c(2n^2 + 3n + 1)$$

Hence,

$$\boxed{n^3 \neq O\big(f(n)\big)}$$

✅ **True**

Statement	Result
$f(n) = O(n^2)$	✅ True
$f(n) = O(n^3)$	✅ True
$n^2 = O\big(f(n)\big)$	✅ True
$f(n) \neq O(n)$	✅ True
$n^3 \neq O\big(f(n)\big)$	✅ True

Check your progress2

Example2: For the function defined by

$f(n) = 2n^3 + 3n^2 + 1 \ \ and \ g(n) = 2n^2 + 3$

Prove/Disprove the following:

$(i) f(n) = \Omega(g(n))$

$(ii)\ g(n) \neq \Omega(f(n))$

$(iii)\ n^3 = \Omega\big(g(n)\big)$

$(iv)\ f(n) \neq \Omega\big(n^4\big)$

$(v)\ n^2 \neq \Omega(f(n))$

Check your progress3:

Example3: For the function defined by

$f(n) = 2n^3 + 3n^2 + 1$ and $g(n) = 2n^3 + 1$,

show that

$(i) f(n) = \Theta(g(n))$

$(ii)\ f(n) \neq \Theta(n^2)$

$(iii)\ n^4 \neq \Theta(g(n))$

Whenever you have a polynomial:

$$a_k n^k + a_{k-1} n^{k-1} + \cdots + a_0$$

the **highest-degree term dominates**.

Here,

$$f(n) = 2n^3 + 3n^2 + 1$$

and

$$g(n) = 2n^3 + 1$$

Both have the highest power n^3, so

$$f(n) = \Theta(n^3), g(n) = \Theta(n^3)$$

Hence,

$$\boxed{f(n) = \Theta\big(g(n)\big)}$$

and any function with a different highest power (such as n^2or n^4) cannot be in the same Θ-class.

Shortcut trick

Look only at the highest powers:

$$f(n) = 2n^3 + 3n^2 + 1 \Rightarrow \Theta(n^3)$$
$$g(n) = 2n^3 + 1 \Rightarrow \Theta(n^3)$$

Since both have the same highest power, they belong to the same Theta class:

$$\boxed{f(n) = \Theta\big(g(n)\big) = \Theta(n^3)}$$

But:

$$n^2 \neq \Theta(n^3), n^4 \neq \Theta(n^3)$$

because the powers are different.

Check your progress4

Q.10: Let f(n) and g(n) be two asymptotically positive functions. Prove or disprove the following (using the basic definition of O, Ω and Θ):

$a)\ 4n^2 + 7n + 12 = O(n^2)$

$b)\ logn + \log(logn) = O(logn)$

$c)\ 3n^2 + 7n - 5 = \Theta(n^2)$

$d)\ 2^{n+1} = O(2^n)$

$e)\ \ 2^{2n} = O(2^n)$

$f)\ f(n) = O(g(n))\ implies\ g(n) = O(f(n))$

g) $\max\{f(n), g(n)\} = \Theta(f(n) + g(n))$

$h)\ f(n) = O(g(n))\ implies\ 2^{f(n)} = O(2^{g(n)})$

$i)\ f(n) + g(n) = \Theta(\min(f(n), g(n)))$

$j)\ 33\mathrm{n}^3\ +\ 4\mathrm{n}^2 = \Omega(\mathrm{n}^4)$